

Name:

Question 1

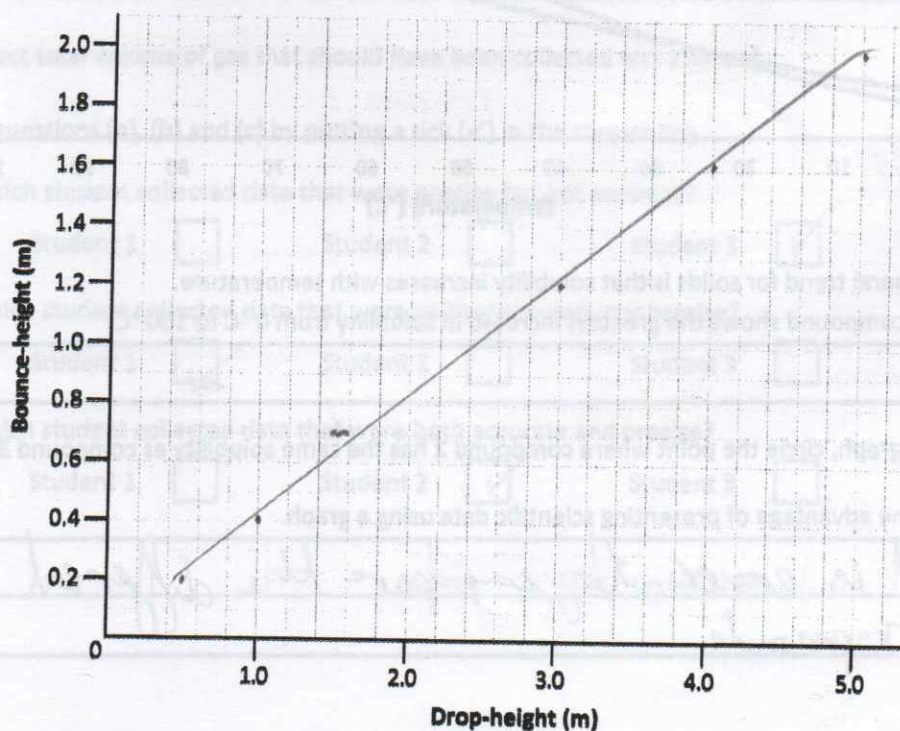
Question 12

(45 marks)

A group of students investigated how high a ball bounces after it is dropped. They allowed a ball to fall from a number of different drop-heights and measured how high the ball bounced each time (the bounce-height). Their results are shown in the table below.

Drop-height (m)	Bounce-height (m)
0.5	0.2
1.0	0.4
1.5	0.6
2.0	0.8
3.0	1.2
4.0	1.6
5.0	2.0

- (a) In the space below, present this information using a suitable graph or chart. (Your graph or chart should allow the viewer to read the results of the investigation and to see any pattern.)



- (b) If the ball was dropped from a height of 10 m, how high would you expect it to bounce?

4 meters

Question 2

Question 3

15 marks

A group of students investigated how solubility in water changes with temperature for solid compounds 1, 2 and 3. The graph below shows the results obtained.

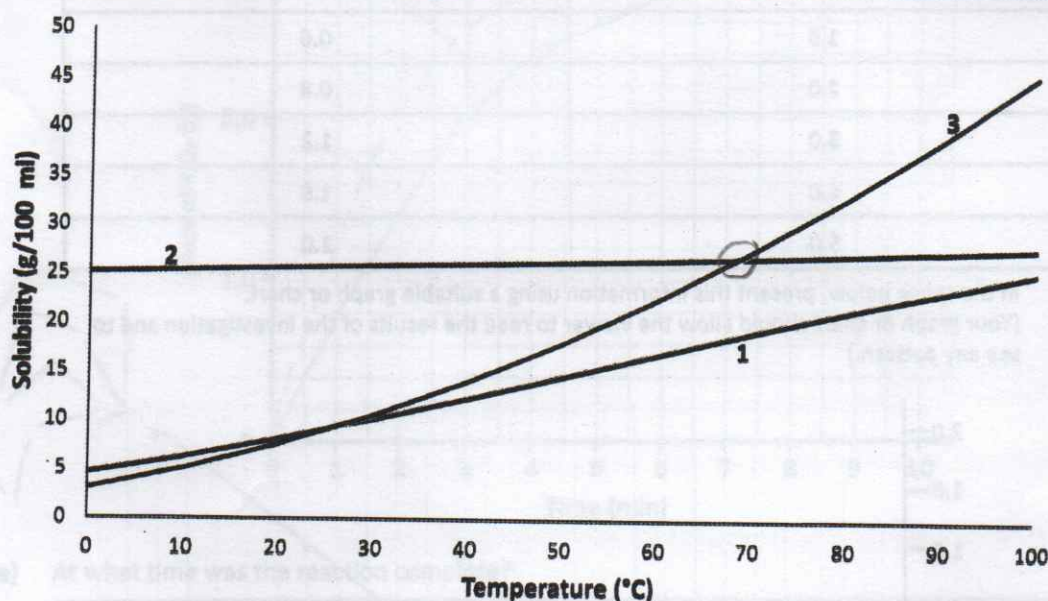
- (a) Hot water was needed during this investigation.

Name an instrument used to heat water in the laboratory.

Bunsen burner

- (b) Describe one safety precaution which should be taken when heating water in the laboratory.

Keep away from electronic equipment.



- (c) The general trend for solids is that solubility increases with temperature.
Which compound shows the greatest increase in solubility from 0 °C to 100 °C?

3

- (d) On the graph, circle the point where compound 2 has the same solubility as compound 3.

- (e) State one advantage of presenting scientific data using a graph.

It is easier to compare the different compounds

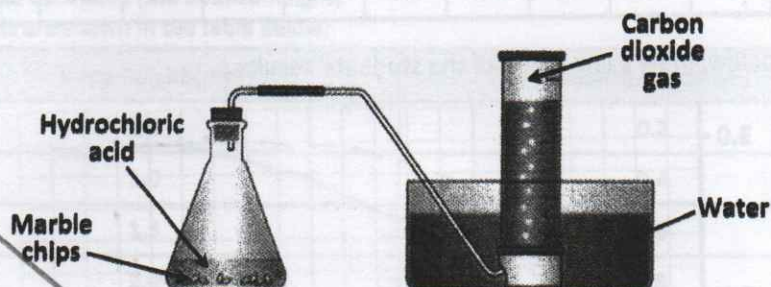
Question 3

Question 12

(30 marks)

Carbon dioxide gas can be produced in the laboratory by reacting hydrochloric acid with marble chips (calcium carbonate).

Three students were asked to measure the total volume of carbon dioxide gas produced when an excess of marble chips was reacted with 100 cm³ of dilute hydrochloric acid. Each student set up the apparatus shown in the diagram. They each repeated the investigation three times. Their results are shown in the table below.



	Run 1	Run 2	Run 3
Student 1	220 cm ³	210 cm ³	230 cm ³
Student 2	240 cm ³	240 cm ³	240 cm ³
Student 3	220 cm ³	220 cm ³	220 cm ³

The correct total volume of gas that should have been collected was 240 cm³.

Answer questions (a), (b) and (c) by putting a tick (✓) in the correct box.

(a) Which student collected data that were precise but not accurate?

Student 1 ☐ Student 2 ☐ Student 3 ☒

(b) Which student collected data that were neither accurate nor precise?

Student 1 ☒ Student 2 ☐ Student 3 ☐

(c) Which student collected data that were both accurate and precise?

Student 1 ☐ Student 2 ☒ Student 3 ☐

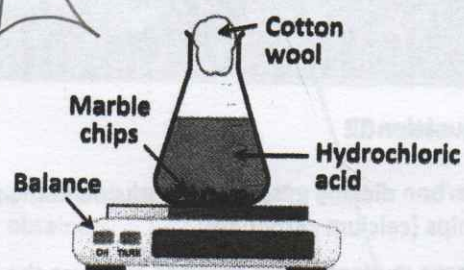
(iv) If the ball was dropped from a height of 10 m, how high would you expect it to bounce?

4 m

The students carried out a second investigation.

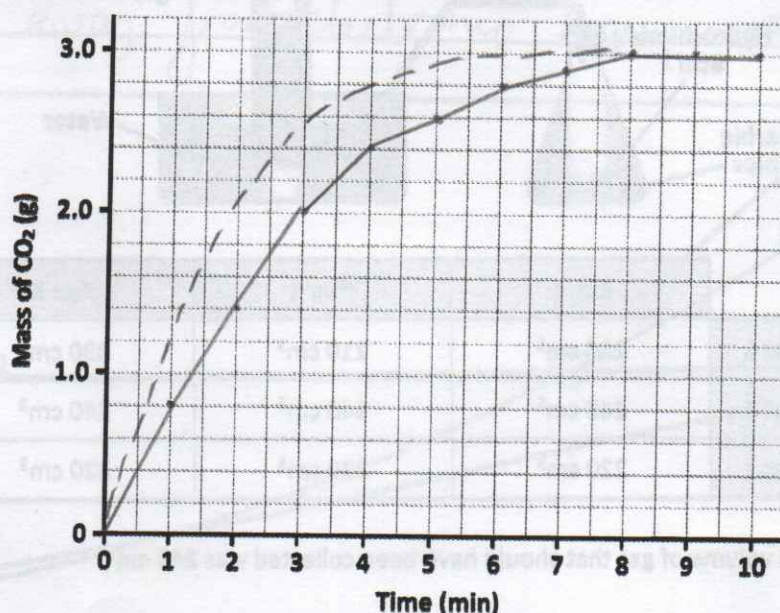
This time they measured the mass of carbon dioxide produced at regular intervals using the apparatus on the right.

The data collected by the students are shown below.



Time (min)	0	1	2	3	4	5	6	7	8	9	10
Mass of CO ₂ (g)	0	0.8	1.4	2.0	2.4	2.6	2.8	2.9	3.0	3.0	3.0

(d) In the space below, draw a line graph of the students' results.



(e) At what time was the reaction complete?

8 minutes

(f) On your graph above, draw a dashed line (-----) to show the result you would expect if the above investigation was repeated exactly as before, but using smaller marble chips of the same total mass.

It is easier to compare the different compounds